Combatting Driver-Specific Overfitting with Image Augmentation

c0: safe driving

c1: texting - right

c2: talking on phone - right

c3: texting - left

c4: talking on phone - left

c5: operating radio

c6: drinking

c7: reaching behind

c8: hair and makeup

c9: talking to passenger

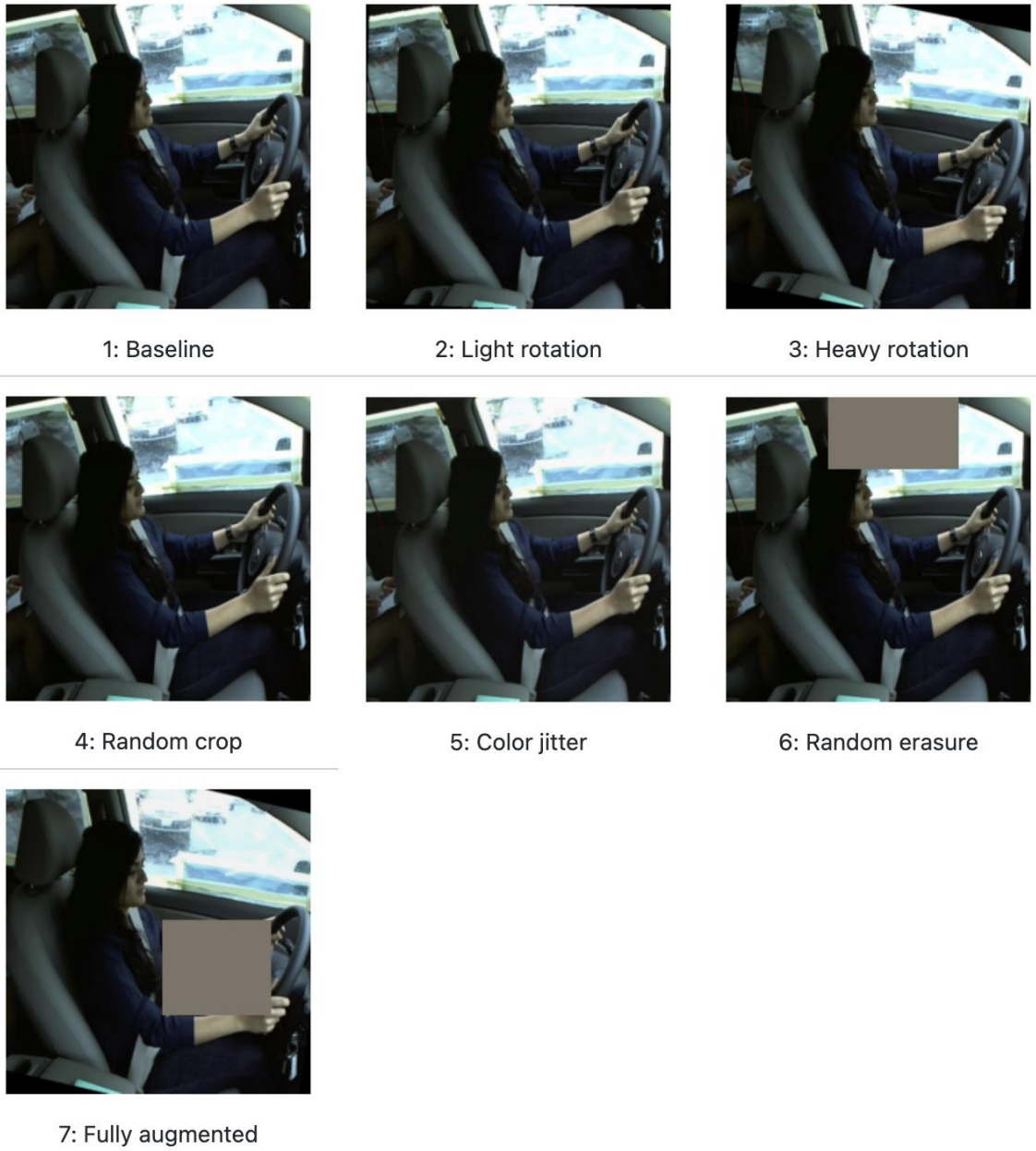
22,424 images, *only 26 individual drivers*

Model **memorizes environments** instead of learning activity signal

Our Approach

ResNet18 model fine-tuned end-to-end,
5-fold grouped CV

One-at-a-time setup with
7 augmentation configurations

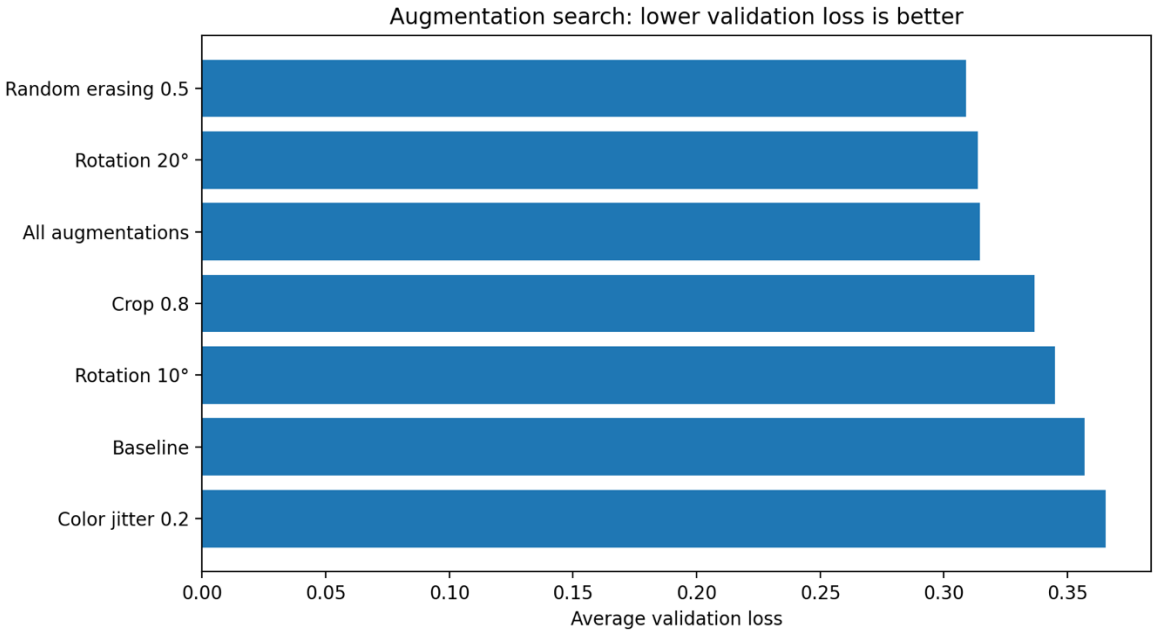


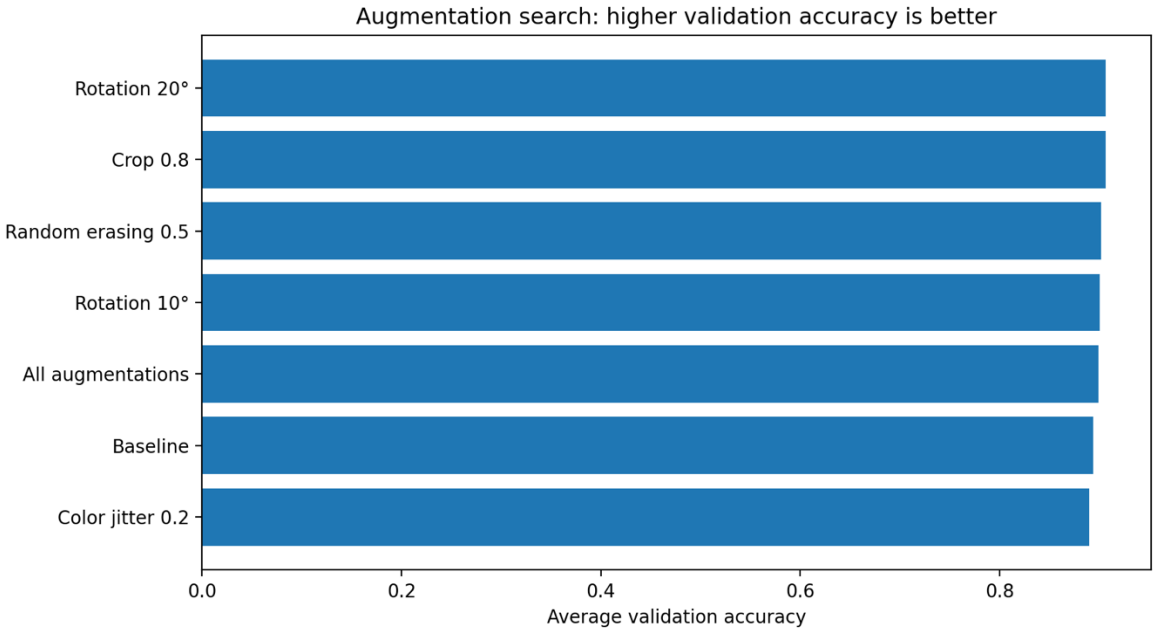
Results

Random erasing and **rotation** most effective

Color jitter provides no benefit
→ Task is *structural* rather then chromatic

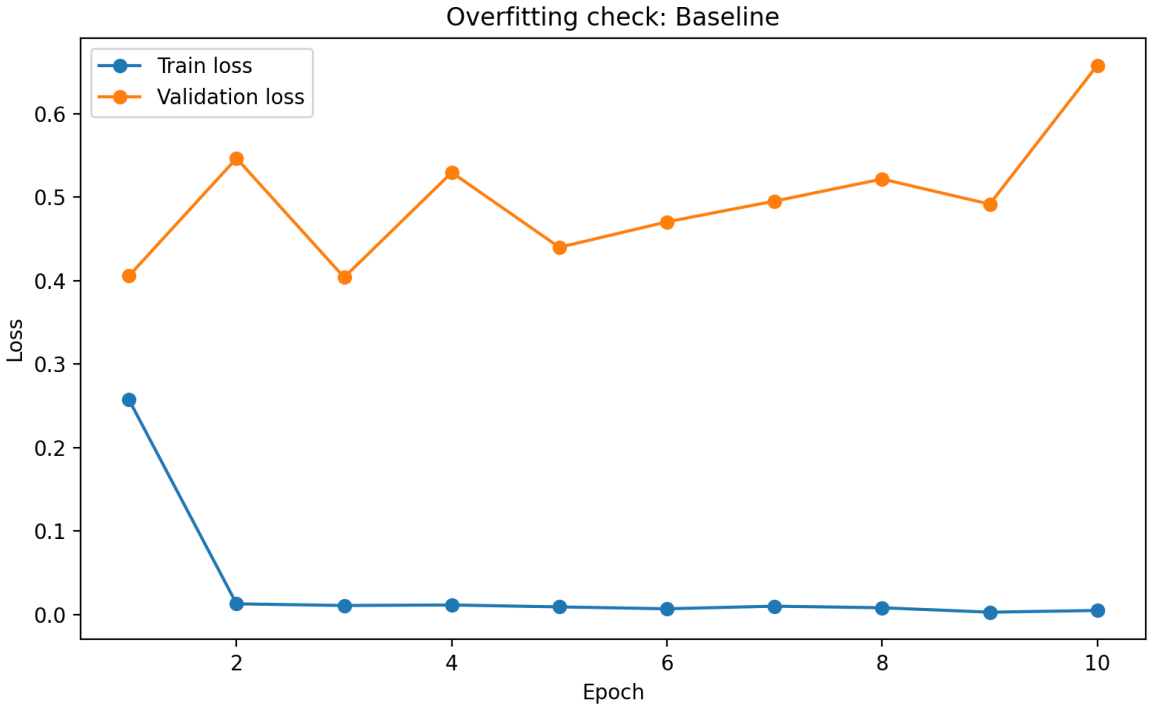
Modest but consistent differences

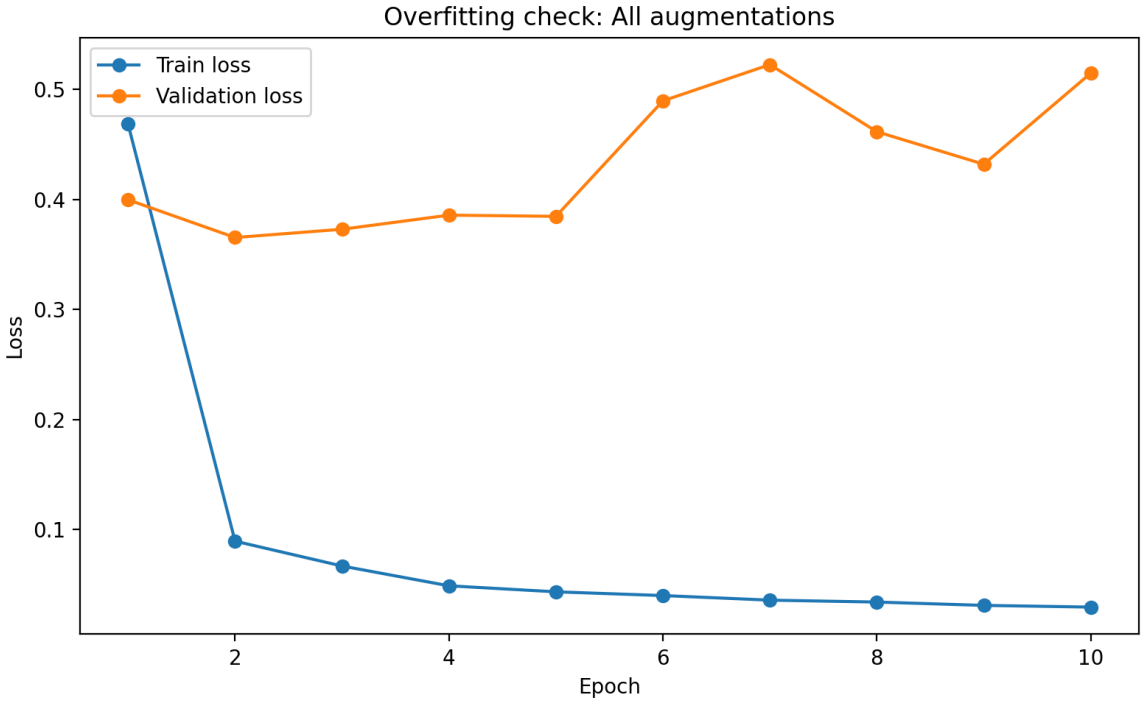




Lessons Learned

Augmentation dampens but does not resolve overfitting
→ Bottleneck is likely *small number of drivers*





Potential Next Steps:

Freezing model backbone

Longer training with early stopping

More folds or repeated CV